MediSync

AI Shift-Handoff & Discharge Copilot

Detailed Project Report

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Contents

1 Problem Statement **2**

1.1 The Healthcare Administrative Crisis . . . 2

1.2 Critical Pain Points . . . 2

1.2.1 Shift Handoffs . . . 2

1.2.2 Discharge Summaries . . . 2

1.2.3 Core Challenges . . . 2

1.3 The Need for Transformation . . . 2

2 Proposed Solution **3**

2.1 MediSync Overview . . . 3

2.2 Solution Architecture . . . 3

2.3 Core Capabilities . . . 3

2.4 Key Differentiators . . . 3

3 Implementation **4**

3.1 System Architecture . . . 4

3.2 Implementation Phases . . . 4

3.2.1 Phase 1: Ingestion Pipeline . . . 4

3.2.2 Phase 2: Memory & Retrieval Pipeline . . . 4

3.2.3 Phase 3: Generation & Copilot UI . . . 5

3.3 Backend & Frontend Implementation Details . . . 6

4 Tech Stack Used **6**

4.1 Frontend Technology Stack . . . 6

4.2 Backend & AI Technology Stack . . . 6

5 Impact **6**

5.1 Clinical & Operational Impact . . . 6

5.2 Key Features for Impact . . . 7

6 Future Scope **7**

6.1 Enhancement Roadmap . . . 7

6.2 Scalability & Compliance . . . 7

7 Conclusion **7**

1 Problem Statement

1.1 The Healthcare Administrative Crisis

Healthcare professionals face a critical challenge: they spend **40-60% of their time** on administrative tasks rather than direct patient care. This administrative burden is particularly acute in two critical workflows that directly impact patient safety and quality of care.

1.2 Critical Pain Points

1.2.1 Shift Handoffs

- When nurses or doctors change shifts, they must communicate patient status, medications, and pending tasks.
- **70% of preventable medical errors** are attributed to poor handoff communications.
- Information is often incomplete, inconsistent, or lost during transitions.
- Manual transcription leads to missed critical details.

1.2.2 Discharge Summaries

- Writing comprehensive discharge documentation takes **30-60 minutes per patient**.
- Summaries are frequently delayed, affecting care continuity.
- Missing critical information leads to hospital readmissions.
- Poor documentation impacts reimbursement and legal records.

1.2.3 Core Challenges

Challenge	Impact
Information Overload	Doctors receive 50+ files per patient (lab reports, imaging, handwritten notes, nursing assessments).
Unstructured Data	Scanned documents, handwritten notes, and mixed-format PDFs are difficult to parse.
Time Poverty	Administrative work cuts into patient interaction time.
Error Risk	Manual transcription leads to missed allergies, medications, or follow-up instructions.

1.3 The Need for Transformation

Healthcare facilities need an intelligent solution that can:

- Automatically process diverse document formats.
- Extract and organize critical patient information.

- Generate accurate, structured clinical documentation.
- Provide source attribution for all generated content.
- Reduce administrative time while improving accuracy.

2 Proposed Solution

2.1 MediSync Overview

MediSync is an AI-powered web application that eliminates clinical administrative burnout by transforming messy, unstructured patient data into perfectly formatted **Discharge Summaries** and **Shift-Handoff Notes** in seconds.

2.2 Solution Architecture

Using advanced OCR (Optical Character Recognition) and **Retrieval-Augmented Generation (RAG)** technology, MediSync enables doctors to:

1. **Upload folders** of scanned handwritten notes, PDFs, and lab reports.
2. **Receive AI-generated documentation** with **clickable citations** linking every claim back to the original source document.
3. **Review, edit, and sign off** on final documentation.

2.3 Core Capabilities

Capability	Description
Automated Ingestion	Drag-and-drop upload of 50+ mixed-format files (PDFs, images, scans).
Intelligent Processing	OCR extracts text from handwritten notes and images; LangChain chunks and embeds content.
Hybrid Search	Combines vector similarity with BM25 keyword search to ensure zero dropped data.
Structured Generation	LLM produces JSON-structured summaries with mandatory fields (Diagnosis, Medications, Allergies, Follow-up, Vitals).
Attribution UI	Every AI-generated claim includes clickable citations linking to the exact source document and page.
Human-in-the-Loop	Doctor reviews, edits, and signs off on final documentation.

2.4 Key Differentiators

- **Clickable Citations**: Every AI-generated claim links back to source documents.
- **Hybrid RAG Pipeline**: Combines vector search with keyword search for maximum accuracy.

- **Structured JSON Output**: Machine-readable clinical data with standardized schema.
- **Privacy-First Design**: PII anonymization before any external API calls.
- **Hospital-Grade UI**: Professional dashboard with medical styling.

3 Implementation

3.1 System Architecture

The system utilizes a modular frontend and backend pipeline. Data flows from React/TypeScript components to a FastAPI backend, through a complex AI processing pipeline involving Text Cleaning, LangChain Chunking, HuggingFace Embeddings, ChromaDB, and a Hybrid Retriever feeding the final LLM Generator.

```
[Frontend (React/TypeScript)] --> [FastAPI Backend] --> [AI Pipeline]
- Upload UI                       - REST APIs           - OCR Engine
- Dashboard                       - Task Queue          - LangChain Chunker
- Review & Citation UI                                  - ChromaDB / BM25
- Export/Sign-off                                       - LLM Generator (JSON)
```

3.2 Implementation Phases

3.2.1 Phase 1: Ingestion Pipeline

Goal: Convert messy, multi-format patient documents into clean, structured text.

Component	Technology	Purpose
File Router	FastAPI	Detect file type, route to appropriate processor
Data Loaders	LangChain	Load CSV, JSON, PDF, text files
Text Cleaner	Regex + spaCy	Remove noise, normalize formatting
Metadata Tagger	Custom	Add source filename, page number, date
PII Shield	Microsoft Presidio	Anonymize patient names, DOBs, IDs

Deliverables:

[✓] CSV/JSON data loader for curated dataset

[✓] File upload endpoint for PDF, PNG, JPG, TIFF

[✓] Text cleaning and normalization

[✓] Metadata extraction (dates, patient IDs)

3.2.2 Phase 2: Memory & Retrieval Pipeline

Goal: Transform clean text into searchable, chunked embeddings stored in ChromaDB.

Component	Technology	Purpose
Text Chunker	LangChain	Split documents into semantic chunks (512-1024 tokens)
Embedding Model	HF all-MiniLM-L6-v2	Generate dense vector representations
Vector Store	ChromaDB	Local persistent storage for embeddings
BM25 Index	rank_bm25	Keyword-based search for exact matches
Hybrid Retriever	LangChain	Combine vector + BM25 using reciprocal rank fusion

Chunking Strategy:

- **Recursive Character Split**: Split on newlines, sentences, paragraphs
- **Chunk Size**: 512-1024 tokens with 20% overlap
- **Metadata**: Preserve source document, page number, chunk index

3.2.3 Phase 3: Generation & Copilot UI

Goal: Generate structured clinical summaries with full source attribution.

LLM-Native Citation Strategy: The system prompt forces the LLM to include `[Chunk_ID]` tags. The frontend parses these tags and creates clickable links to source documents.

JSON Output Schema:

```
{
  "patient_summary": {
    "chief_complaint": "string",
    "diagnosis": ["string"],
    "medications": [
      {"name": "string", "dosage": "string", "frequency": "string", "route": "st
    ],
    "allergies": ["string"],
    "vitals": {
      "blood_pressure": "string", "heart_rate": "string",
      "temperature": "string", "respiratory_rate": "string"
    },
    "follow_up_instructions": "string"
  },
  "citations": [
    {
      "claim": "string", "source_document": "string",
      "page_number": "number", "chunk_id": "string"
    }
  ]
}
```

3.3 Backend & Frontend Implementation Details

The system defines precise API routes (e.g., `/api/v1/upload`, `/api/v1/summary`) mapping to modular Python services (`ingestion.py`, `rag_pipeline.py`, `generation.py`). The frontend leverages `React Router` to manage Dashboard, Upload, Documents, and Summaries states seamlessly.

4 Tech Stack Used

4.1 Frontend Technology Stack

Component	Technology	Purpose
Framework	React + Vite	Production-grade UI framework
Language	TypeScript	Type-safe development
UI Library	Tailwind CSS + shadcn/ui	Hospital dashboard aesthetic
State Management	Zustand	Lightweight state for upload/progress
Data Fetching	TanStack Query	Server state management

4.2 Backend & AI Technology Stack

Component	Technology	Purpose
API Framework	FastAPI	High-performance async REST API
LLM	**Cohere Command-R7B**	Primary: Enterprise-grade generation
Framework	LangChain	Orchestrate RAG pipeline
Embeddings	HF all-MiniLM-L6-v2	Dense vector representations
Vector Store	ChromaDB	Local persistent vector database
Keyword Search	rank_bm25	BM25 for hybrid search
PII Anonymizer	Microsoft Presidio	Auto-mask PII before API calls

5 Impact

5.1 Clinical & Operational Impact

- **Time Savings**: Reduce documentation time from 30-60 min to under 5 min per patient.
- **Error Reduction**: Eliminate missed allergies, medications through automated extraction.
- **Care Continuity**: Enable faster, more accurate handoffs between shifts.
- **Readmission Reduction**: Improve discharge summary quality to reduce preventable readmissions.
- **Workflow Efficiency**: Automated document processing replaces manual data entry.

5.2 Key Features for Impact

1. **Clickable Citations**: Build trust through transparency - doctors can verify every AI-generated claim.
2. **Human-in-the-Loop**: Final review ensures accuracy before signing.
3. **Dual Output**: Supports both Discharge Summaries and Shift Handoff Notes.
4. **Privacy First**: PII anonymization protects patient data before external processing.

6 Future Scope

6.1 Enhancement Roadmap

Timeline	Planned Features
Short-Term (3-6 mo)	OCR Integration (GPT-4o/Claude Vision), Real-time Collaboration, EMR Integration, Voice Input Dictation.
Medium-Term (6-12 mo)	Multi-language Support, Specialty Templates (Cardiology, Pediatrics), Quality Metrics, Audit Logging.
Long-Term (1-2 yrs)	**ABDM Integration**, SNOMED CT Codes auto-mapping, Predictive Analytics, Telehealth Integration, Voice AI Assistant.

6.2 Scalability & Compliance

- **Cloud Deployment**: Move from local ChromaDB to cloud vector databases.
- **Multi-tenant Architecture**: Support multiple hospitals/clinics with Redis caching.
- **HIPAA Compliance**: Ensure all PHI handling meets US healthcare standards.
- **GDPR Ready**: Support European data protection requirements (ISO 27001).

7 Conclusion

MediSync represents a significant step forward in healthcare documentation automation. By combining cutting-edge AI technologies (RAG, LLMs, embeddings) with a user-friendly interface, the platform addresses the critical administrative burden facing healthcare providers.

The project's hybrid architecture balances cost-effectiveness (local GPU embeddings) with enterprise-grade capabilities (Cohere LLM), making it accessible for healthcare facilities of various sizes.

With the foundation now in place, MediSync is well-positioned to expand into a comprehensive clinical documentation platform that can integrate with existing healthcare infrastructure and evolve with the advancing AI landscape.